

Calculator Test Cases

TC 001 - Addition of Two Positive Numbers

Test Description

Verify that calculator adds two positive numbers correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 10
2. Press +
3. Enter 5
4. Press =

Expected Result

Calculator should display 15.

TC 002 - Addition of Negative Numbers

Test Description

Verify addition works with negative numbers.

Preconditions

Calculator application is opened.

Test Steps

1. Enter -8
2. Press +
3. Enter -2
4. Press =

Expected Result

Calculator should display -10.

TC 003 - Addition of Decimal Numbers

Test Description

Verify calculator adds decimal numbers correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 5.5
2. Press +
3. Enter 2.3
4. Press =

Expected Result

Calculator should display 7.8.

TC 004 - Subtraction of Two Numbers

Test Description

Verify subtraction works correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 20
2. Press -
3. Enter 8
4. Press =

Expected Result

Calculator should display 12.

TC 005 - Negative Result in Subtraction

Test Description

Verify calculator shows negative result correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 5
2. Press -
3. Enter 10
4. Press =

Expected Result

Calculator should display -5.

TC 006 - Multiplication by Zero

Test Description

Verify multiplication with zero works correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 25
2. Press *
3. Enter 0
4. Press =

Expected Result

Calculator should display 0.

TC 007 - Multiplication of Decimal Numbers

Test Description

Verify multiplication of decimal values.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 2.5
2. Press *
3. Enter 4
4. Press =

Expected Result

Calculator should display 10.

TC 008 - Division of Two Numbers

Test Description

Verify division works correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 20
2. Press /
3. Enter 4
4. Press =

Expected Result

Calculator should display 5.

TC 009 - Division by Zero

Test Description

Verify calculator handles division by zero.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 10
2. Press /
3. Enter 0
4. Press =

Expected Result

Calculator should display an error message Or can't divide by zero.

TC 010 - Invalid Character Input

Test Description

Verify calculator rejects alphabetic input.

Preconditions

Calculator application is opened.

Test Steps

1. Enter abc
2. Press =

Expected Result

Calculator should display invalid input message.

TC 011 - Empty Input Validation

Test Description

Verify calculator handles empty input correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Leave input field empty
2. Press =

Expected Result

Calculator should display input required message.

TC 012 - Multiple Operators Together

Test Description

Verify calculator handles multiple operators properly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 10
2. Press +
3. Press *
4. Press =

Expected Result

Calculator should display invalid expression message.

TC 013 - BODMAS Rule Verification

Test Description

Verify calculator follows BODMAS rule correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter $10 + 5 * 2$
2. Press =

Expected Result

Calculator should display 20.

TC 014 - Large Number Addition

Test Description

Verify calculator handles large numbers correctly.

Preconditions

Calculator application is opened.

Test Steps

1. Enter 999999
2. Press +
3. Enter 111111
4. Press =

Expected Result

Calculator should display 1111110.

TC 015 - Negative Number Multiplication

Test Description

Verify multiplication of negative numbers.

Preconditions

Calculator application is opened.

Test Steps

1. Enter -5
2. Press *
3. Enter 2
4. Press =

Expected Result

Calculator should display -10.